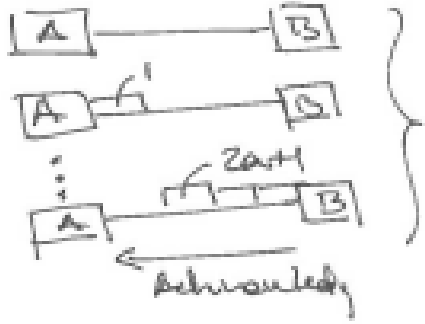
# E3.17 Communications Network Answers – 2023

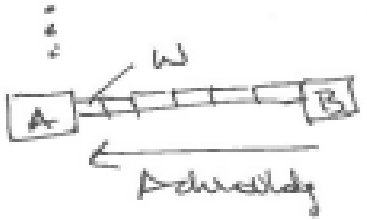
## Q1

Q1) a) Timing diagram can be used

i) $W < 2a+1$




$W > 2a+1$




[4]

ii) Each error generates K retransmission

$$N_R = \sum_{i=1}^{\infty} f(i)\, P^{i-1}(1-P) \qquad f(i) = 1+(i-1)K$$

$$= \frac{1-P+KP}{1-P}$$

$$U_{(W>2a+1)} = \frac{1-P}{1+2aP}$$

assume $K \sim 2a+1$

$$U_{(W<2a+1)} = \frac{W(1-P)}{(1+2a)(1-P+WP)}$$

assume $K \sim W$

[6]

Q1

b) i) SNMP is an application layer protocol to convey management control and information messages between managing server and agents
- Request-response mode
- Unsolicited message are also permitted
SNMP lies on top of UDP [3]

ii) ICMP is used on hosts and routers to communicate network-layer information e.g. error reporting
- messages are carried inside IP datagrams i.e. as a payload
- The structure of the message is similar to the way TCP/UDP payloads are passed to upper layers. [3]

iii) when a sender wants to send data RTS
The AP node broadcast a CTS message instructing other stations not to send for the reserved duration
Benefits
- The hidden station problem is mitigated
- A collision involving RTS or CTS will have a very short duration
But
RTS/CTS introduces delay and consume channel resources
Suggestion is to use only to transmit long DATA. [4]

Q2

Q2

a)

i) Link-state algorithm:

- Use of complete knowledge about the network
- Requires a mechanism to continuously update the state of the network
- Solution can be performed in one node or in multiple nodes

[4]

ii) distance vector algorithm:

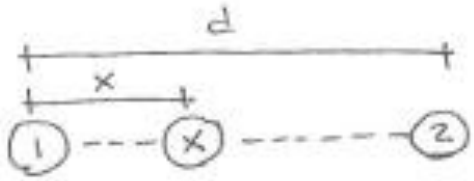
- Calculate the shortest path in an iterative and distributed manner.
- Each node starts with the knowledge of its attached links only
- As neighbours exchange information the routing converges to the shortest path.

| | | | |
|---|---|---|---|
| RIP | IGP | Distance-Vector | UDP |
| OSPF | IGP | Link-state | IP |
| BGP | EGP | Path-Vector | TCP |

[4]

Q2

b)




$d = 20\ m$

$x = 0 \quad e_t + e_0 + e_1 d^3 = 10\times10^{-10} + 8\times10^{3}\times10^{-10}$

$\approx 8\times10^{-7}$

$x > 0 \Rightarrow$ 2 hops

$10\times10^{-10} + 10^{-10}(20-x)^3 + 10^{-10} + 10^{-10}x^3$

$\frac{\partial}{\partial x}\left(20\times10^{-10} + 10^{-10}(20-x)^3 + 10^{-10}x^3\right) = 0$

$-3(20-x)^2 + 3x^2 = 0$

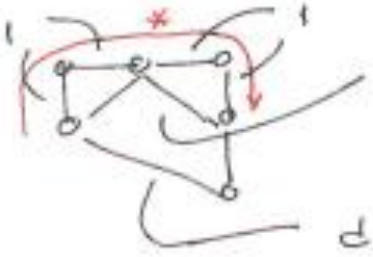
$(20-x)^2 = x^2$

$20 - x = x \qquad \boxed{x = 10}$

$20\times10^{-10} + 2\times10^{-10}(10)^3$

$\approx 2\times10^{-7}$ [6]

ii)




$d = \sqrt{1+1} \quad - \quad d^3 = (\sqrt{2})^3 = 2.82$

$d = \sqrt{2^2+1} \quad - \quad d^3 = (\sqrt{5})^3 = 11.18$ [3]

* "shortest" path

iii)



● = NODES

discussion req i) & ii) [3]

Q3

Q3.1 5

i) Scheduling: it place packets according to a predefined policy.

A default scheduling is normally set as first in first out (FIFO): packet are forward in the order that they arrive to the queue

Another example is weighted fair queeps. extended the diversion.

Queue management: example of managed tools would be: remove, discard, remark packets if e.g. queue is approaching a pre-agreed threshold.
The aim is to proactively avoiding the queue overflowing [3]

ii) The rate at which tokens are generated + diversion [3]

iii) The size of the bucket + diversion [3]

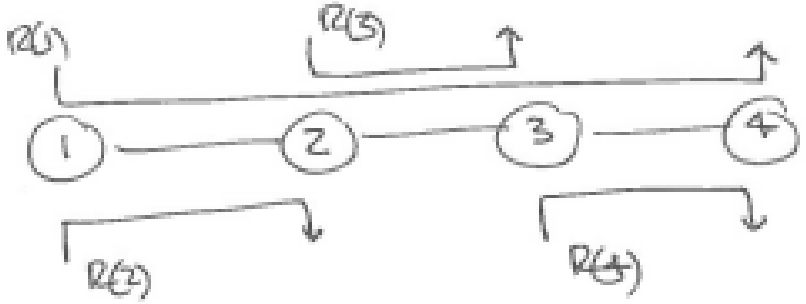
iv) Jitter is the variation in delay between transmission and reception over a network connection.

It can be dealt with by placing a client-side buffering and implementing an adaptive playout delay. [3]

Q3z

After reading the note the network is:

i)




$R(2) = R(3) = R(4)$

$R(1) = 1 - R(2)$

$\max \sum_{i=1}^{4} w(i) \ln R(i)$

$\max \left( w(1) \ln R(1) + 3 w(2) \ln (1 - R(1)) \right)$

$\frac{\partial}{\partial R(1)} \left( w(1) \ln R(1) + 3 w(2) \ln (1 - R(1)) \right) = 0$

$\frac{w(1)}{R(1)} = \frac{3 w(2)}{1 - R(1)}$

$R(1) = \frac{w(1)}{w(1) + 3 w(2)}$ $\quad R(2) = R(3) = R(4) = \frac{3 w(2)}{w(1) + 3 w(2)}$

ii) if $w(i) = 1$ for all $i$ then we have implemented the proportional fair flow assignment. [3]

$R(1) = \frac{1}{4} \quad R(2) = R(3) = R(4) = \frac{3}{4}$ [2]

iii) if $w(1) = 1$ and $w(2) = \frac{1}{3}$ we have implemented the max-min flow assignment.

$R(1) = \frac{1}{2} \quad R(2) = R(3) = R(4) = \frac{1}{2}$ [3]

Q4 Q4.1

The slow start state balance the amount of data a sender can transmit. (congestion window) with the capacity to receive data (receiver window). the lower of the two values is the maximum amount of data that can be transmitted.

If the slow start state is currently operating at congestion window = 1 it doubles the congestion window at every acknowledgement received upto a predefined threshold.

Congestion avoidance state continues to probe the network for additional capacity but slowly (linear rate). That is less aggressively than in the slow start state. [5]

Additive Increase / multiplicative decrease

Increase the transmission rate (window size) probing for more capacity (bandwidth) until loss occurs.

When a loss is detected the transmitter decreases the transmission rate by a multiplicative factor e.g. half the window size before detection of loss. [2]

Q4.2

i) From data given

$C = 4$ Kbits/s $\quad \frac{1}{\mu} = 6$ Kbit/packet

$\mu_i = \mu C = \frac{4}{6}$ packets/second

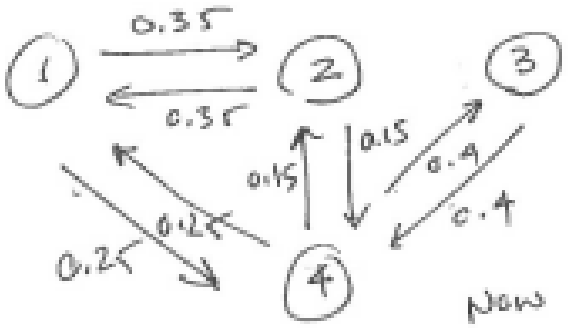
$\gamma_T = \sum \frac{\lambda_i}{\mu C_i - \lambda_i} = N$ $\quad \gamma_i$ in packet/second

There is only one possible way of arcing traffic and the network load is symetric

$$N = 2\ \frac{0.6}{0.\overline{6} - 0.6} + 2\ \frac{0.5}{0.\overline{6} - 0.5} + 2\ \frac{0.4}{0.\overline{6} - 0.4}$$ [4]

ii) The policy is to place a bidirectional link between nodes that have the highest demand That is nodes ① ↔ ② = 0.35 packet/sec the network now looks like (and gives assignment using min hop shortest path.) [3]

iii)




Now bottleneck link is between nodes ③ & ④ [3]

iv)

N' for resulting network



$$2 \times \frac{0.35}{0.\overline{6} - 0.35} + 2 \times \frac{0.25}{0.\overline{6} - 0.25} + 2 \times \frac{0.15}{0.\overline{6} - 0.15} +$$

$$2 \times \frac{0.4}{0.\overline{6} - 0.4}$$

[3]

Compare N & N'